

Practical 12 - Non-linear effects and splines - Stata

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Datasets required: `SAheart.csv`

In this practical we will explore modelling non-linear covariate effects using the same South Africa data as the previous two practicals.

Question 1

Load the data, and fit a simple logistic model for `chd` with `sbp` as the single covariate, saving the model for later comparison:

```
logit chd sbp
est store linear
```

What do the model results show?

Question 2

Generate fitted probability values from the model in question and plot these against `sbp`. Can you explain what you see?

Question 3

Use the following code to plot a lowess smoother of `chd` against `sbp`.

```
lowess chd sbp
```

What does the lowess smoother suggest about the effect of SBP on CHD?

Question 4

Re-run the `lowess` function with the `logit` option. What has changed compared to the previous plot?

Question 5

Fit a logistic regression for the `chd` outcome with `sbp` and the square of `sbp` as covariates. To do this you can generate a new variable in the data frame for the square term and include it in the model, or you can directly include the quadratic term with `c.sbp#c.sbp`.

From your fitted model what do you conclude now about the effect of SBP?

Use a plot to compare the fitted log odds values from this model with the ones estimated using loess. (Hint: you can use `lowess` as a plot type within the `twoway` function)

Question 6

Use the following code to fit a logistic model for `chd` with a restricted cubic spline with 5 knots for the effect of `sbp`:

```
makespline rcs sbp, knots(5) replace  
logit chd _rs_rcs_1 _rcs*
```

Then create a plot to compare the fitted log odds with the loess and quadratic model estimates.

Question 7

Re-run your code from question 6 but with either smaller values of `knots` or larger values. Does changing the value of `knots` change the fitted log odds in the way you would expect?

Question 8

Use the `lrtest` function to compare your earlier quadratic effect model with the restricted cubic spline model with `knots(5)`. What do you conclude?

Is it valid to compare these two models with a likelihood ratio test?

Use `estimates stats` function to compare the two models with the AIC.

Question 9

We have throughout this practical implicitly taken the loess smoother estimate as if were ‘correct’. It is though just an estimate, and in particular the estimate it gives is affected by the bandwidth, which is specified by the `span` argument to `loess`. Examine the documentation for `loess` and re-run it with different values of `span` to see how the value affects the loess estimate of the probability of CHD by SBP.